

ConvLSTM-GCN-transformer: Spatiotemporal graph-attention model for vegetation index map forecasting

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ARTICLE INFO

Keywords:

Deep learning
Vegetation health forecasting
Graph neural networks
Transformer
Remote sensing

ABSTRACT

Vegetation indices (VIs) are fundamental tools for assessing vegetation health, productivity, and ecosystem changes under growing climate and land-use pressures. Reliable forecasting of these indices is essential for sustainable resource management and climate adaptation. However, accurate VI prediction remains challenging due to the nonlinear, nonstationary, and spatiotemporal nature of vegetation dynamics. Most existing deep learning models predict single-point or regional-average values, neglecting spatial continuity and inter-pixel dependencies needed for accurate map-based forecasting. Conversely, methods that generate spatially explicit VI maps often perform inconsistently across heterogeneous landscapes that combine forests, water bodies, urban zones, and bare soil. These challenges highlight the need for architectures that integrate local and global spatial context while preserving temporal coherence across diverse environments. This paper introduces a hybrid ConvLSTM-GCN-Transformer architecture for high-resolution VI forecasting from time-series data. ConvLSTM captures temporal evolution, the Graph Convolutional Network (GCN) models local spatial dependencies, and the Transformer enhances global feature representation. Trained on monthly NDVI maps from Landsat imagery (1996–2025) over northern Tunisia, the model achieved an RMSE of 0.034 and an NSE of 0.866, outperforming baseline models. Two generalization studies were conducted to evaluate robustness: (i) cross-sensor evaluation using Sentinel-2 and MODIS imagery (RMSEs of 0.064 and 0.09), and (ii) cross-index adaptation to the Enhanced Vegetation Index (EVI) and Soil-Adjusted Vegetation Index (SAVI). All codes are publicly available to support reproducibility ([Link here](#)).

1. Introduction

The combined effect of climate change and land-use transformation are exerting increasing pressures on ecosystems, thereby positioning vegetation monitoring as a critical priority for sustainable development (Wen et al., 2025). Vegetation indices (VIs) provide quantitative measures of vegetation health, biomass, and productivity, serving as essential tools for assessing ecosystem dynamics (Kooistra et al., 2024). Derived from multispectral satellite imagery, VIs are widely used for drought assessment, crop yield estimation, and land degradation analysis (Ferchichi et al., 2022). As global climate variability increases, accurate and timely VI prediction has become fundamental for sustainable land management and food security (Sishodia et al., 2020; Zeng et al., 2022). Reliable forecasting supports proactive decision-making, optimizes resource allocation, and strengthens adaptive strategies that enhance ecosystem resilience.

Modeling vegetation dynamics has evolved significantly over the past decade, moving from classical statistical techniques to modern deep learning (DL) frameworks (Liu et al., 2024). Early statistical approaches such as autoregressive (AR), ARIMA, and seasonal ARIMA models (Fernández-Manso et al., 2011; Mohanty et al., 2025) captured seasonal trends but struggled with the nonlinear and nonstationary behavior of satellite-derived vegetation data (Mohanty et al., 2025). Machine learning (ML) algorithms such as Random Forests (RF), Support Vector Machines (SVM), and Multilayer Perceptrons (MLPs) later introduced data-driven flexibility, improving predictive accuracy (Ang et al., 2022; Khaki & Wang, 2019). However, these models typically treat spatial observations as independent, neglecting spatial and temporal dependencies that govern vegetation growth and degradation (Balti et al., 2023; Wang et al., 2025).

Deep learning has further advanced vegetation modeling through architectures capable of learning sequential patterns and complex

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<https://doi.org/10.1016/j.eswa.2026.131596>

Received 28 November 2025; Received in revised form 1 February 2026; Accepted 7 February 2026

Available online 12 February 2026

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dependencies. Convolutional Neural Networks (CNNs), Long Short-Term Memory networks (LSTMs), and hybrid CNN–LSTM frameworks have shown strong performance in capturing temporal and contextual dynamics (Gao et al., 2023; Reddy & Prasad, 2018). Yet, most of these approaches predict single or regional-average values, overlooking fine-scale spatial details and continuity required for high-resolution mapping.

Recent studies have begun shifting toward map-based prediction frameworks that forecast full-resolution VI maps instead of point-based estimates (Schwalbert et al., 2018). ConvLSTM architectures have demonstrated the ability to model spatiotemporal dependencies and capture seasonal NDVI variations (Ahmad et al., 2023; Kartal et al., 2024; Robin et al., 2022). More advanced frameworks such as Graph WaveNet, GWConvLSTM, and ConvLSTM-SAC-NL have incorporated graph structures or attention mechanisms to improve the representation of spatial relationships and enhance computational efficiency (Beyer et al., 2023; Cai et al., 2024; Xu et al., 2024). However, the existing methods are developed on small or spatially uniform datasets and generalize poorly to large, heterogeneous regions. This limitation underscores the need for models capable of integrating both local and global spatial information while maintaining temporal coherence across diverse landscapes.

Recent advances in Graph Convolutional Networks (GCNs) and Transformers offer promising solutions to these challenges. GCNs model spatial dependencies by representing data as graphs, propagating features across neighboring nodes to maintain local coherence. This approach has proven effective in applications such as traffic prediction and remote sensing (Bhatti et al., 2023; Jiang et al., 2019; Wu et al., 2020). Transformers, initially designed for natural language processing (Vaswani et al., 2017), employ self-attention to capture long-range dependencies and have since been adapted for vision tasks such as land-cover mapping and environmental monitoring (Dosovitskiy, 2020; Tang et al., 2023). Their integration into environmental modeling has shown strong potential in medical imaging, land-cover mapping, and urban planning (Chen et al., 2021). The complementary strengths of GCNs local structural learning and Transformers in global contextual reasoning make their combination particularly suitable for spatially explicit vegetation forecasting (Dong et al., 2021; Huo et al., 2023; Šajina et al., 2025).

The current VI prediction literature reveals three key gaps. First, most studies predict single or regional-average VI values, disregarding spatial patterns and dependencies that are essential for accurate map-based forecasting. Second, existing map-based approaches capture spatial variability but show weak generalization across sensors or diverse landscapes such as forests, wetlands, and urban zones. Third, research has mainly focused on NDVI, with limited attention to other vegetation indices such as EVI or SAVI, thereby reducing model adaptability in broader monitoring contexts. These limitations emphasize the need for an architecture that can jointly model temporal, spatial, and multi-index dynamics using large, heterogeneous satellite datasets.

This paper introduces a ConvLSTM–GCN–Transformer architecture for high-resolution VI maps prediction. The model processes historical VI time series data to produce full-resolution monthly forecasts across diverse landscapes. ConvLSTM layers capture temporal evolution, GCN modules enforce spatial consistency through pixel adjacency graphs, and Transformer encoders extract long-range contextual patterns. To evaluate robustness, two generalization experiments were performed: (i) cross-sensor evaluation using Sentinel-2 and MODIS imagery, and (ii) cross-index adaptation to EVI and SAVI.

The key contributions of this study are summarized as follows:

- A hybrid ConvLSTM–GCN–Transformer architecture integrating spatial structures and temporal dynamics for VI map prediction.
- A graph-based spatial formulation using sparse GCN operations to efficiently model spatial interactions over large-scale satellite data.

- The model is trained on a heterogeneous landscape encompassing forests, urban areas, water bodies, and bare soil over an extended temporal range (1996–2025), enabling robust learning across diverse environmental conditions.
- Validation of the architecture’s adaptability through rigorous cross-sensor generalization (Sentinel-2, MODIS) and cross-index generalization (EVI, SAVI).

The remainder of this paper is structured as follows: [Section 2](#) reviews the state-of-the-art in vegetation index forecasting. [Section 3](#) introduces the proposed methodology and model architecture. [Section 4](#) describes the datasets and the preprocessing procedures. [Section 5](#) presents the experimental framework, results, ablation studies, and generalization experiments. [Section 6](#) provides an in-depth discussion of the findings. [Section 7](#) examines the limitations and proposes future research directions. Finally, [Section 8](#) concludes the paper, summarizing the key contributions and findings.

2. Related works

This section reviews key developments in VI prediction and advanced deep learning models relevant to spatiotemporal forecasting. It outlines major approaches, from traditional scalar-based prediction to recent map-based methods, emphasizing their strengths and current limitations. Finally, it discusses the emerging use of graph and Transformer architectures in other domains, which inspires the design of our proposed hybrid framework.

2.1. Previous works on VI prediction

Research in VIs prediction primarily follows two paradigms: scalar prediction, which estimates average index values over points or regions, and map-based prediction, which produces spatially detailed maps. These approaches differ in methods, data needs, and applications. This section reviews the main contributions, methods, and limitations in both areas and identifies gaps that motivate more advanced map-based models.

Scalar prediction of VIs, such as NDVI, focuses on forecasting average values for specific points or regions, inherently overlooking spatial variability. Early methods employed statistical and machine learning techniques. (Fernández-Manso et al., 2011) applied Seasonal ARIMA (SARIMA) models to predict 10-day NDVI values using NOAA AVHRR data (1993–1997) in Spain, capturing seasonal trends but struggling with non-linear dynamics, resulting in limited accuracy for complex vegetation scenarios. Liang et al. (2024) applied Support Vector Machines (SVM) and Random Forests (RF) for satellite data prediction, leveraging high-dimensional data but failing to model temporal dependencies, thereby restricting dynamic monitoring applications.

Recurrent neural networks, especially Long Short-Term Memory (LSTM) models (Hochreiter & Schmidhuber, 1997), improved temporal modeling. Reddy and Prasad (2018) predicted MODIS-based NDVI (2000–2016) in the Great Nicobar Islands with an RMSE below 0.03, but only regional averages were considered. Vasilakos et al. (2022) combined Dynamic Time Warping (DTW) similarity with LSTM to forecast NDVI on Lesvos Island, Greece, achieving RMSEs between 0.017 and 0.079. Gao et al. (2023) developed TSD-CNN-LSTM, decomposing NDVI into trend, seasonal, and residual components for enhanced temporal accuracy. More recently, Dey et al. (2025) proposed an LSTM using Sentinel-2 and meteorological data, improving performance by 4–8% over classical models, but still limited to point-based predictions. Scalar methods are efficient but discard spatial patterns, motivating pixel-level approaches for detailed tasks like plot monitoring or land degradation assessment (Zeng et al., 2022).

Map-based prediction has emerged to overcome scalar method limitations by generating spatially explicit maps. Initial efforts employed Convolutional Long Short-Term Memory (ConvLSTM) architectures,

blending convolutional operations with LSTM units to model spatiotemporal dynamics. Robin et al. (2022) predicted high-resolution NDVI maps across Africa with Sentinel-2, achieving RMSE of 0.1882 and NSE of 0.0270, but faced computational limits and struggled with long-range dependencies. Ahmad et al. (2023) achieved RMSE of 0.0782 with a ConvLSTM model, but convolution's local receptive field limited performance. To improve accuracy, hybrid architectures emerged. Beyer et al. (2023) introduced a Graph WaveNet (GWNN) model, surpassing ConvLSTM by 31% in RMSE with faster training, though graph construction is computationally expensive. Xu et al. (2024) proposed ConvLSTM-SAC-NL with spatial autocorrelation and non-local attention, achieving RMSE of 0.061 and NSE of 0.045 at 1 km MODIS resolution, but being sensitive to noise. Cai et al. (2024) introduced GWConvLSTM, combining spatial weights with ConvLSTM, effective when spatial weights are high quality. Yu et al. (2021) applied LSTM-GRUs for short-term VI maps, limited in modeling complex spatial interactions. Berger et al. (2019) proposed a probabilistic graphical model (PCGM) for NDVI prediction, with an RMSE of 0.0989, but less accurate than modern deep learning methods.

Despite advances, map-based methods still face many challenges. ConvLSTM captures local spatiotemporal patterns but struggles with long-range dependencies and high-resolution data. Graph-based models like GWNN need precise graphs, limiting their generalization. Most studies focus on NDVI, with few addressing SAVI or EVI. Dependence on specific sensors, such as Sentinel-2 or MODIS, reduces adaptability. These challenges motivate hybrid models that combine local and global modeling, handle noise, and generalize across indices and sensors.

Table 1 summarizes key studies on VIs prediction, highlighting the used methods and findings.

2.2. SOTA on hybrid graph and transformer models

Graph Convolutional Networks (GCNs) have become a cornerstone in computer science for modeling non-Euclidean data structures, particularly in spatiotemporal domains where entities exhibit relational dependencies. GCNs operate by propagating features across graph nodes through convolution-like operations, enabling the capture of local spatial relationships (Kipf, 2016; Zhao et al., 2019). In remote sensing and traffic prediction, GCNs have been employed to represent sensor networks or pixel adjacencies as graphs, improving feature aggregation in heterogeneous environments (Beyer et al., 2023; Yu et al., 2020; Zhao et al., 2019).

Transformers, originally designed for natural language processing (Vaswani et al., 2017), have revolutionized sequence modeling in computer science through self-attention mechanisms, which compute global dependencies without recurrence. In spatiotemporal tasks, Transformers excel at handling long-range interactions, making them suitable for large-scale data like satellite imagery (Cong et al., 2022; Yuan & Lin, 2020). Montero et al. (2022) applied Transformers for spectral index computation in remote sensing, highlighting their ability to process high-dimensional multispectral data efficiently.

Recent advancements have explored hybrid GCN-Transformer models, combining GCN's local graph modeling with Transformer's global attention for improved spatiotemporal reasoning. Šajina et al. (2025) proposed GCN-Transformer, a model for multi-person pose forecasting that integrates Graph Convolutional Networks with Transformers, achieving superior performance in capturing both local joint relations and global motion patterns. Similarly, Li and Ma (2022) developed a hybrid model based on improved Transformer and GCN for COVID-19 forecasting, reporting enhanced accuracy (RMSE reductions up to 15%) by leveraging GCN for spatial dependency modeling and Transformer for temporal sequence handling in epidemiological data. In air quality prediction, Raj et al. (2025) introduced a hybrid graph convolutional LSTM model incorporating Transformer elements, demonstrating RMSE improvements of 10–20% over baselines by fusing GCN's spatial graphs with Transformer's attention for multivariate time series. Additionally, Gao et al. (2024) proposed hybrid graph transformer networks for

multivariate time series anomaly detection, achieving F1-scores above 0.9 by combining GCN's node-level convolutions with Transformer's self-attention for robust detection in complex networks. These hybrids demonstrate superior handling of complex dependencies in large datasets, with faster convergence and reduced computational overhead compared to pure models, as evidenced by up to 30% efficiency gains in training time.

In the field of VIs map prediction, such hybrid approaches hold significant promise. By leveraging GCN for pixel-level spatial coherence through graph-based propagation that enforces smoothness across neighboring pixels and Transformer for long-range patterns, these models enable more accurate, scalable VI forecasting.

3. Methodology

This section introduces the methodology for developing and evaluating the proposed architecture that integrates Convolutional LSTM, Graph Convolutional Networks, and attention modules to model complex spatiotemporal dynamics in sequential image data. This architecture, illustrated in Fig. 1, leverages the complementary strengths of local temporal encoding, spatial graph-based reasoning, and global contextual modeling to achieve high-fidelity predictions.

3.1. Overall pipeline description

The proposed architecture, detailed in Algorithm 1, processes a sequence of images through a series of carefully designed stages to produce a spatially explicit prediction. Given a sequence of single-channel image observations, $\mathcal{X} = \{X_t\}_{t=1}^T$ where each $X_t \in \mathbb{R}^{H \times W \times 1}$, represents a single-channel image at time t , the objective is to predict the next image $Y \in \mathbb{R}^{H \times W \times 1}$.

The full input tensor is $\mathcal{X} \in \mathbb{R}^{B \times T \times H \times W \times 1}$, where B is the batch size and T is the number of time steps, and (H, W) represents the spatial dimensions of each single-channel image. The pipeline begins with a ConvLSTM2D layer that captures temporal dynamics while preserving spatial structure, producing a feature tensor $\mathbf{X}_{\text{lstm}} \in \mathbb{R}^{B \times H \times W \times C}$, where C is the number of feature channels.

$$\mathbf{X}_{\text{lstm}} = \text{ConvLSTM2D}(\mathcal{X}) \quad (1)$$

This is followed by batch normalization to stabilize training, yielding $\mathbf{X}_{\text{norm}} \in \mathbb{R}^{B \times H \times W \times C}$, which mitigates internal covariate shift and enhances convergence.

$$\mathbf{X}_{\text{norm}} = \text{BatchNorm}(\mathbf{X}_{\text{lstm}}) \quad (2)$$

To enable graph-based processing, the spatial grid is flattened into a node representation $\mathbf{X}_{\text{nodes}} \in \mathbb{R}^{B \times N \times C}$, where $N = H \times W$ represents the number of nodes in the graph.

A GCN module then propagates information across a sparse adjacency matrix encoding local pixel connectivity, producing $\mathbf{X}_{\text{gcn}} \in \mathbb{R}^{B \times N \times D}$.

$$\mathbf{X}_{\text{gcn}} = \text{GcnBlock}(\mathbf{X}_{\text{norm}}) \quad (3)$$

Subsequently, a Transformer Encoder captures global dependencies across all nodes, resulting in $\mathbf{X}_{\text{tr}} \in \mathbb{R}^{B \times N \times D}$.

$$\mathbf{X}_{\text{tr}} = \text{TransformerBlock}(\mathbf{X}_{\text{gcn}}) \quad (4)$$

Finally, the node features are reshaped back to the image format $\mathbf{X}_{\text{img}} \in \mathbb{R}^{B \times H \times W \times D}$, and a pointwise 2D convolution followed by a sigmoid activation generates the predicted output $\hat{\mathbf{Y}} \in \mathbb{R}^{B \times H \times W \times 1}$.

$$\mathbf{X}_{\text{img}} = \text{Reshape}(\mathbf{X}_{\text{tr}}) \quad (5)$$

$$\hat{\mathbf{Y}} = \sigma(\text{Conv2D}(\mathbf{X}_{\text{img}})) \in \mathbb{R}^{H \times W \times 1} \quad (6)$$

where σ denotes the sigmoid activation function, applied to ensure that the output values are constrained within the $[0, 1]$ interval, which is suitable for image-based prediction tasks.

Table 1
Summary of state-of-the-art vegetation index prediction methods.

Reference	Output Format	Method	Description
Gao et al. (2023)	Average values	TSD-CNN-LSTM	Time-series decomposition with CNN-LSTM for NDVI prediction
Reddy and Prasad (2018)	Average values	LSTM	MODIS NDVI forecasting using LSTM networks
Dey et al. (2025)	Average values	LSTM	Sentinel-2 and meteorological data integration for NDVI prediction
Vasilakos et al. (2022)	Average values	DTW- LSTM	Fuzzy clustering with LSTM for MODIS NDVI forecasting
Fernández-Manso et al. (2011)	Average values	Seasonal ARIMA	SARIMA model for 10-day NDVI prediction using AVHRR data
Robin et al. (2022))	Image/Map	ConvLSTM	High-resolution NDVI map prediction across Africa
Robin et al. (2022)	Image/Map	Graph WaveNet	Graph-based spatiotemporal modeling for NDVI maps
Ahmad et al. (2023)	Image/Map	ConvLSTM	ConvLSTM architecture for NDVI map forecasting
Xu et al. (2024)	Image/Map	ConvLSTM-SAC-NL	Spatial autocorrelation and non-local attention integration
Cai et al. (2024)	Image/Map	GWConv-LSTM	Geographically weighted ConvLSTM for NDVI prediction
Yu et al. (2021)	Image/Map	LSTM-GRUs	RNN-based approach for short-term vegetation index maps
Berger et al. (2019)	Image/Map	PCGM	Probabilistic graphical model for NDVI map prediction

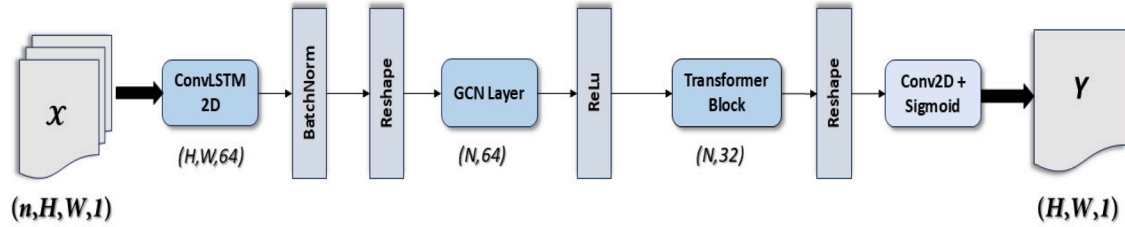


Fig. 1. Global architecture of the proposed ConvLSTM-GCN-Transformer model.

Algorithm 1 provides a step-by-step description of this pipeline, and **Fig. 1** illustrates the flow of data through each processing module.

Algorithm 1 ConvLSTM-GCN-transformer forward pipeline.

Require: A temporal sequence $\mathcal{X} \in \mathbb{R}^{B \times T \times H \times W \times 1}$

Ensure: Predicted image $\hat{\mathbf{Y}} \in \mathbb{R}^{B \times H \times W \times 1}$

```

1: for  $b = 1$  to  $B$  do
2:   Apply temporal encoding:
3:    $\mathbf{X}_{\text{lstn}}^{(b)} \leftarrow \text{ConvLSTM2D}(\mathcal{X}^{(b)}) \in \mathbb{R}^{H \times W \times C}$ 
4:   Apply batch normalization:
5:    $\mathbf{X}_{\text{norm}}^{(b)} \leftarrow \text{BatchNorm}(\mathbf{X}_{\text{lstn}}^{(b)})$ 
6:   Flatten spatial dimensions into nodes:
7:    $\mathbf{X}_{\text{nodes}}^{(b)} \in \mathbb{R}^{N \times C}$ , with  $N = H \times W$ 
8:   Apply GCN layer:
9:    $\mathbf{X}_{\text{gcn}}^{(b)} \leftarrow \text{GcnBlock}(\mathbf{X}_{\text{nodes}}^{(b)})$ 
10:  Apply Transformer encoder:
11:   $\mathbf{X}_{\text{tr}}^{(b)} \leftarrow \text{TransformerBlock}(\mathbf{X}_{\text{gcn}}^{(b)}) \in \mathbb{R}^{N \times D}$ 
12:  Reshape to image format:  $\mathbf{X}_{\text{img}}^{(b)} \in \mathbb{R}^{H \times W \times D}$ 
13:  Apply final convolution:
14:   $\hat{\mathbf{Y}}^{(b)} \leftarrow \sigma(\text{Conv2D}(\mathbf{X}_{\text{img}}^{(b)})) \in \mathbb{R}^{H \times W \times 1}$ 
15: end for
16: Stack all outputs  $\hat{\mathbf{Y}}^{(b)}$ 
17: return  $\hat{\mathbf{Y}} \in \mathbb{R}^{B \times H \times W \times 1}$ 

```

3.2. ConvLSTM block

LSTM networks were designed to overcome the vanishing gradient problem in recurrent neural networks, enabling the modeling of long-range temporal dependencies in sequential data (Hochreiter & Schmidhuber, 1997). However, standard LSTM models operate on one-dimensional vectorized inputs, limiting their ability to exploit spatial structure inherent in image sequences. ConvLSTM was introduced by Shi et al. (2015) as an extension of the LSTM architecture that integrates convolutional operations into the input-to-state and state-to-state transitions.

The ConvLSTM block (Fig. 2), is designed to capture both temporal dynamics and local spatial relationships within an input sequence $\mathbf{X} \in \mathbb{R}^{B \times T \times H \times W \times 1}$. Unlike traditional LSTM units that operate on vec-

torized data, the ConvLSTM2D architecture employs convolutional operations, which preserve spatial structures while modeling temporal dependencies.

This capability is particularly important for processing image sequences, where both spatial coherence and temporal evolution are critical. The block processes each time step t of the input sequence and produces a spatiotemporal feature tensor $\mathbf{X}_{\text{lstn}} \in \mathbb{R}^{B \times H \times W \times C}$, where C denotes the number of output channels determined by the convolutional filters.

For each time step t , the ConvLSTM updates the **hidden state** $\mathbf{H}_t \in \mathbb{R}^{B \times H \times W \times C}$ and **cell state** $\mathbf{C}_t \in \mathbb{R}^{B \times H \times W \times C}$ using the following equations:

$$i_t = \sigma(W_{xi} * \mathbf{X}_t + W_{hi} * \mathbf{H}_{t-1} + b_i) \quad (7)$$

$$f_t = \sigma(W_{xf} * \mathbf{X}_t + W_{hf} * \mathbf{H}_{t-1} + b_f) \quad (8)$$

$$g_t = \tanh(W_{xg} * \mathbf{X}_t + W_{hg} * \mathbf{H}_{t-1} + b_g) \quad (9)$$

$$o_t = \sigma(W_{xo} * \mathbf{X}_t + W_{ho} * \mathbf{H}_{t-1} + b_o) \quad (10)$$

where $*$ denotes the convolution operation, $W_{xi}, W_{hi}, W_{xf}, W_{hf}, W_{xg}, W_{hg}, W_{xo}, W_{ho} \in \mathbb{R}^{k \times k \times C}$ are convolutional kernels with kernel size k , σ is the sigmoid activation function, $\tanh$ is the hyperbolic tangent and $b_i, b_f, b_g, b_o \in \mathbb{R}^C$ are the corresponding biases.

The cell state and hidden state are updated as:

$$\mathbf{C}_t = f_t \odot \mathbf{C}_{t-1} + i_t \odot g_t \quad (11)$$

$$\mathbf{H}_t = o_t \odot \tanh(\mathbf{C}_t) \quad (12)$$

where $\odot$ denotes element-wise multiplication. The final output of the ConvLSTM block is taken as the hidden state at the last time step:

$$\mathbf{X}_{\text{lstn}} = \mathbf{H}_T \quad (13)$$

Using convolutions instead of fully connected layers ensures the preservation of spatial locality, as each gate applies learnable filters to its input and the previous hidden state. This design significantly reduces the number of trainable parameters compared to standard LSTMs and improves computational efficiency, especially when processing high-resolution images. The use of a 3×3 kernel achieves a balance between capturing local spatial patterns and maintaining computational efficiency.

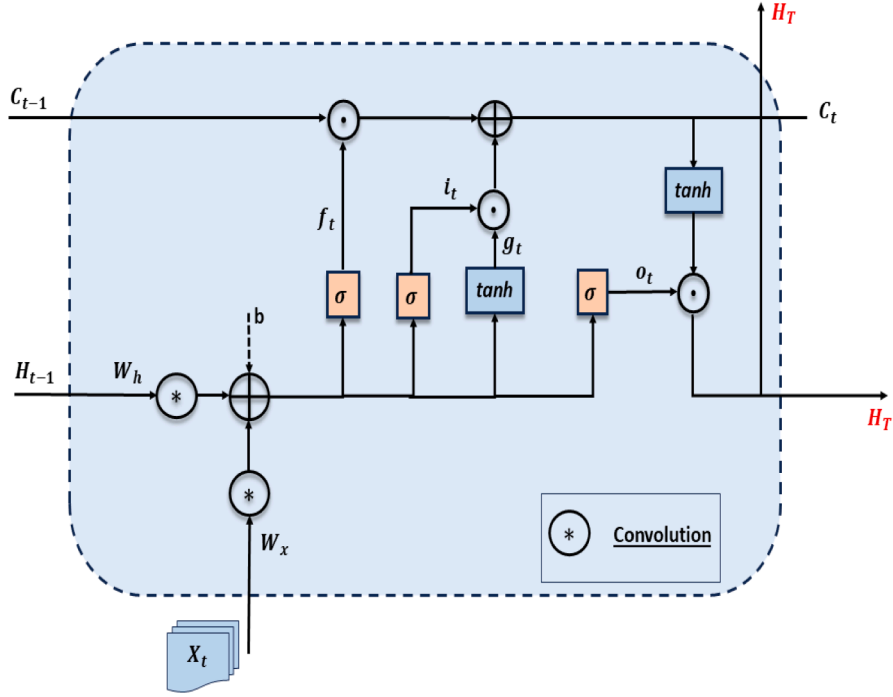


Fig. 2. ConvLSTM2D block architecture.

Algorithm 2 summarizes the ConvLSTM processing steps, and Fig. 2 illustrates the convolutional operations applied within the recurrent structure.

Algorithm 2 ConvLSTM processing.

Require: Temporal sequence $X \in \mathbb{R}^{B \times T \times H \times W \times 1}$

Ensure: ConvLSTM features $X_{\text{lstn}} \in \mathbb{R}^{B \times H \times W \times C}$

```

1: Initialize  $H_0 \leftarrow 0, C_0 \leftarrow 0$ 
2: for  $b = 1$  to  $B$  do
3:   for  $t = 1$  to  $T$  do
4:     Compute gates:
5:      $i_t \leftarrow \sigma(W_{xi} * X_t^{(b)} + W_{hi} * H_{t-1}^{(b)} + b_i)$ 
6:      $f_t \leftarrow \sigma(W_{xf} * X_t^{(b)} + W_{hf} * H_{t-1}^{(b)} + b_f)$ 
7:      $g_t \leftarrow \tanh(W_{xg} * X_t^{(b)} + W_{hg} * H_{t-1}^{(b)} + b_g)$ 
8:      $o_t \leftarrow \sigma(W_{xo} * X_t^{(b)} + W_{ho} * H_{t-1}^{(b)} + b_o)$ 
9:     Update cell state:
10:     $C_t^{(b)} \leftarrow f_t \odot C_{t-1}^{(b)} + i_t \odot g_t$ 
11:    Update hidden state:
12:     $H_t^{(b)} \leftarrow o_t \odot \tanh(C_t^{(b)})$ 
13:   end for
14:   Set  $X_{\text{lstn}}^{(b)} \leftarrow H_T^{(b)}$ 
15: end for
16: return  $X_{\text{lstn}} \in \mathbb{R}^{B \times H \times W \times C}$ 

```

3.3. GCN Block

Graph Convolutional Networks (GCNs) extend convolutional neural networks to non-Euclidean domains, enabling learning on graph-structured data (van den et al., 2017). This innovation opened new avenues for spatial modeling in contexts where data points have complex topological relationships, such as social networks, molecular structures, or remote sensing grids (Bhatti et al., 2023).

The GCN block (Algorithm 3), models local spatial relationships by representing the input image as a graph, where each pixel is a node, and edges connect each pixel to its four immediate neighbors (top, bottom, left, right). This 4-neighbor topology is encoded in a sparse adjacency

matrix $A_{\text{sparse}} \in \mathbb{R}^{N \times N}$, where $N = H \times W$. The matrix contains non-zero entries only for adjacent pixels, significantly reducing the computational complexity to $\mathcal{O}(N)$ for matrix operations, compared to $\mathcal{O}(N^2)$ for dense matrices. This sparse representation effectively preserves local spatial coherence, which is crucial for maintaining smooth transitions in the predicted output.

Given a batch of node embeddings $X \in \mathbb{R}^{B \times N \times C}$, the graph convolution operation applies the following:

$$X_{\text{gcn}} = \sigma(A_{\text{sparse}} X W_{\text{gcn}} + b_{\text{gcn}}) \quad (14)$$

Where $W_{\text{gcn}} \in \mathbb{R}^{C \times D}$ is a learnable weight matrix, $b_{\text{gcn}} \in \mathbb{R}^D$ is a learnable bias vector, and σ is the ReLU activation function.

The adjacency matrix A_{sparse} enables information propagation among neighboring nodes, thus aggregating local features and enhancing spatial consistency in the learned representation.

To ensure numerical stability and account for variability in node connectivity, the adjacency matrix is normalized using the symmetric normalization scheme:

$$A_{\text{sparse}} = D^{-1/2} A D^{-1/2} \quad (15)$$

where D is the diagonal degree matrix defined as $D_{ii} = \sum_j A_{ij}$. This normalization prevents nodes with a high degree from dominating the feature aggregation process, thereby stabilizing the gradient flow and improving model convergence.

The spatial graph employs a 4-neighbor adjacency scheme, where connections are restricted to orthogonal neighbors. This design choice is motivated by both theoretical and practical considerations: at 30m resolution, local spatial continuity in agricultural and natural landscapes is primarily governed by direct adjacency relationships, while diagonal connections introduce computational redundancy. Graph densification (e.g., 8-neighbor connectivity) would increase edge count by a factor of approximately two, escalating memory and computational overhead for processing large satellite scenes. This sparse connectivity provides a balanced representation of local spatial structure while maintaining computational tractability for regional-scale analysis.

Algorithm 3 summarizes the computational steps of the GCN block, and Fig. 3 provides a visual representation of the graph convolution process.

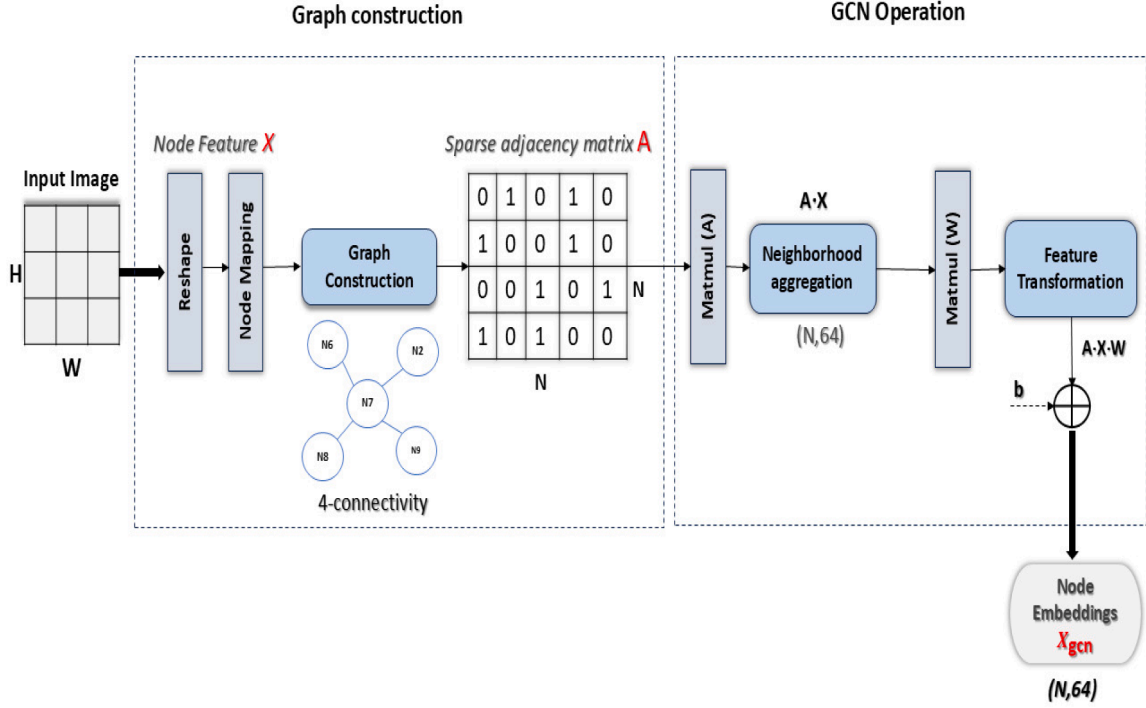


Fig. 3. GCN block architecture showing graph construction and convolution operations.

Algorithm 3 GCN processing.

Require: Normalized feature map $\mathbf{X}_{\text{norm}} \in \mathbb{R}^{B \times H \times W \times C}$

Ensure: GCN features $\mathbf{X}_{\text{gcn}} \in \mathbb{R}^{B \times N \times D}$

```

1:  $N \leftarrow H \times W$ 
2: for  $b = 1$  to  $B$  do
3:   Flatten  $\mathbf{X}_{\text{norm}}^{(b)}$  into  $\mathbf{X}_{\text{nodes}}^{(b)} \in \mathbb{R}^{N \times C}$ 
4:   for each pixel  $i$  do
5:     Connect  $i$  to top, bottom, left, and right neighbors
6:   end for
7:   Store edges in  $\mathbf{A}_{\text{sparse}} \in \mathbb{R}^{N \times N}$ 
8: end for
9: for  $b = 1$  to  $B$  do
10:  Compute:  $\mathbf{X}_{\text{gcn}}^{(b)} \leftarrow \sigma(\mathbf{A}_{\text{sparse}} \mathbf{X}_{\text{nodes}}^{(b)} \mathbf{W}_{\text{gcn}} + \mathbf{b}_{\text{gcn}})$ 
11: end for
12: return  $\mathbf{X}_{\text{gcn}} \in \mathbb{R}^{B \times N \times D}$ 

```

3.4. Transformer block

The Transformer architecture, initially proposed by Vaswani et al. (2017) for NLP, introduced the self-attention mechanism, allowing models to capture long-range dependencies without relying on recurrent structures. Its flexibility and effectiveness have since led to widespread adoption in vision tasks, including image classification, segmentation and spatiotemporal prediction (Radionov et al., 2017; Wang et al., 2017).

The attention block, illustrated in Fig. 4, is designed to capture global contextual dependencies across all spatial positions, thereby complementing the local spatial modeling performed by the GCN. Given the GCN output $\mathbf{X}_{\text{gcn}} \in \mathbb{R}^{B \times N \times D}$, the Transformer employs a multi-head self-attention mechanism to model long-range interactions among the nodes.

For each sample in the batch, the input is first projected into query, key, and value representations:

$$\mathbf{Q} = \mathbf{X}_{\text{gcn}} \mathbf{W}_Q, \quad \mathbf{K} = \mathbf{X}_{\text{gcn}} \mathbf{W}_K, \quad \mathbf{V} = \mathbf{X}_{\text{gcn}} \mathbf{W}_V \quad (16)$$

Where $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V \in \mathbb{R}^{D \times d}$ are learnable weight matrices, $d = D/h$ is the dimension of each attention head and h is the number of attention heads.

This decomposition enables efficient parallel computation of attention across multiple subspaces. For each attention head i , the self-attention is computed as:

$$\mathbf{A}^{(i)} = \text{softmax}\left(\frac{\mathbf{Q}^{(i)} \mathbf{K}^{(i)\top}}{\sqrt{d}}\right), \quad \mathbf{O}^{(i)} = \mathbf{A}^{(i)} \mathbf{V}^{(i)} \quad (17)$$

Where $\mathbf{A}^{(i)} \in \mathbb{R}^{N \times N}$ is the attention weight matrix for head i and $\mathbf{O}^{(i)} \in \mathbb{R}^{N \times d}$ is the corresponding output.

The outputs from all heads are concatenated and linearly projected to the original embedding dimension:

$$\mathbf{Z} = \text{Concat}(\mathbf{O}^{(1)}, \dots, \mathbf{O}^{(h)}) \mathbf{W}_O, \quad \mathbf{W}_O \in \mathbb{R}^{D \times D} \quad (18)$$

A residual connection and layer normalization are then applied to stabilize training:

$$\mathbf{X}_1 = \text{LayerNorm}(\mathbf{X}_{\text{gcn}} + \text{Dropout}(\mathbf{Z})) \quad (19)$$

The intermediate features $\mathbf{X}_1$ are further refined using a **position-wise feed-forward network (FFN)**:

$$\text{FFN}(\mathbf{X}_1) = \max(0, \mathbf{X}_1 \mathbf{W}_1 + \mathbf{b}_1) \mathbf{W}_2 + \mathbf{b}_2 \quad (20)$$

Where $\mathbf{W}_1 \in \mathbb{R}^{D \times D_{\text{ff}}}$, $\mathbf{b}_1 \in \mathbb{R}^{D_{\text{ff}}}$, $\mathbf{W}_2 \in \mathbb{R}^{D_{\text{ff}} \times D}$, $\mathbf{b}_2 \in \mathbb{R}^D$, and D_{ff} is the dimensionality of the hidden layer in the FFN.

A final residual connection and normalization complete the block:

$$\mathbf{X}_{\text{tr}} = \text{LayerNorm}(\mathbf{X}_1 + \text{Dropout}(\text{FFN}(\mathbf{X}_1))) \quad (21)$$

This Transformer module employs multi-head self-attention to capture long-range dependencies across all spatial positions. The attention mechanism operates on the node embeddings $\mathbf{X}_{\text{gcn}} \in \mathbb{R}^{B \times N \times D}$ produced by the preceding GCN block, where $D = 32$ is the compact embedding dimension. While the number of tokens $N = H \times W$ remains large, the reduced embedding dimension D significantly lowers the per-token computational cost of the attention operations. Furthermore, the implementation leverages memory-efficient algorithms available in modern deep

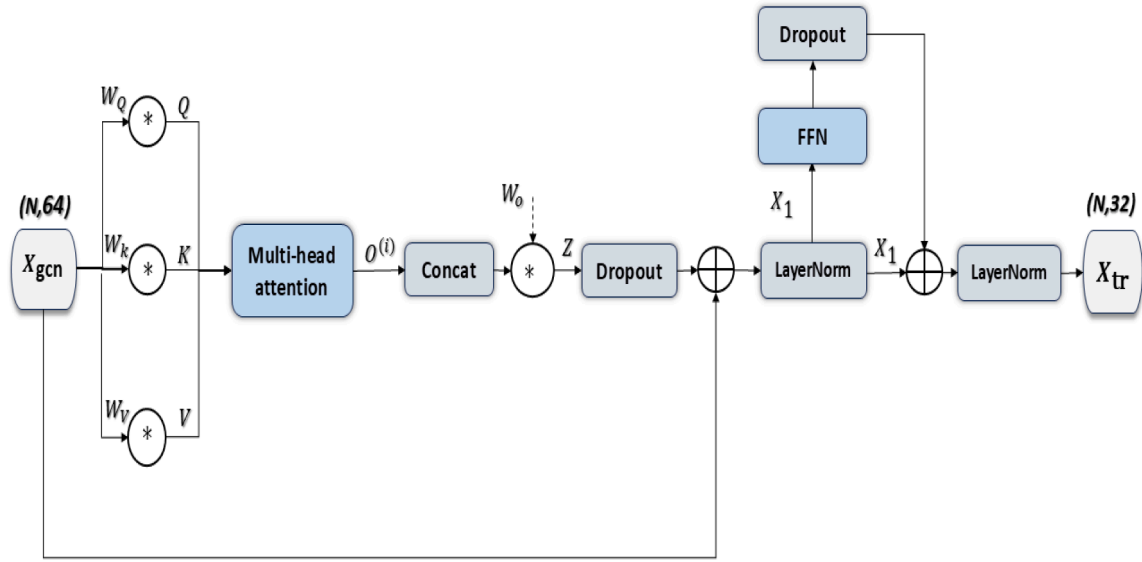


Fig. 4. Transformer block architecture with multi-head attention and feed-forward network.

learning frameworks. TensorFlow's MultiHeadAttention layer employs optimizations such as chunked computation and gradient checkpointing when processing long sequences, strategies that maintain the global receptive field of standard self-attention while reducing peak memory consumption. This enables processing of the full token set without requiring explicit downsampling, pooling, or custom sparse attention patterns.

Although explicit positional encoding is commonly employed in Transformer architectures, spatial positional information in the proposed framework is implicitly preserved by earlier modules. The ConvLSTM maintains absolute spatial locality through its convolutional operations on the 2D image grid, while the GCN explicitly encodes pixel adjacency via a fixed 4-neighbor graph. Consequently, the node embeddings provided to the Transformer already encapsulate rich structural and topological information, enabling the Transformer to effectively model global relationships without reconstructing spatial layout from scratch.

This architecture empowers the model to capture complex and large-scale spatial relationships. The use of multiple attention heads enhances the robustness to diverse spatial interactions, while residual connections and normalization layers contribute to stable and efficient training.

Algorithm 4 outlines the full Transformer pipeline, and Fig. 4 provides a visual overview of the attention and FFN components.

Algorithm 4 Transformer encoder block.

Require: GCN output $\mathbf{X}_{\text{gcn}} \in \mathbb{R}^{B \times N \times D}$

Ensure: Transformer features $\mathbf{X}_{\text{tr}} \in \mathbb{R}^{B \times N \times D}$

```

1: for  $b = 1$  to  $B$  do
2:   Compute  $\mathbf{Q}, \mathbf{K}, \mathbf{V}$  from  $\mathbf{X}_{\text{gcn}}^{(b)}$ 
3:   for each head  $i = 1$  to  $h$  do
4:      $\mathbf{A}^{(i)} \leftarrow \text{softmax}\left(\frac{\mathbf{Q}^{(i)} \mathbf{K}^{(i)\top}}{\sqrt{d}}\right)$ 
5:      $\mathbf{Z}^{(i)} \leftarrow \mathbf{A}^{(i)} \mathbf{V}^{(i)}$ 
6:   end for
7:   Concatenate:  $\mathbf{Z} \leftarrow \text{Concat}(\mathbf{Z}^{(1)}, \dots, \mathbf{Z}^{(h)}) \mathbf{W}_O$ 
8:   Residual connection:  $\mathbf{X}_1 \leftarrow \text{LayerNorm}(\mathbf{X}_{\text{gcn}}^{(b)} + \text{Dropout}(\mathbf{Z}))$ 
9:   Feed-forward:  $\text{FFN}(\mathbf{X}_1) \leftarrow \max(0, \mathbf{X}_1 \mathbf{W}_1 + \mathbf{b}_1) \mathbf{W}_2 + \mathbf{b}_2$ 
10:   $\mathbf{X}_{\text{tr}}^{(b)} \leftarrow \text{LayerNorm}(\mathbf{X}_1 + \text{Dropout}(\text{FFN}(\mathbf{X}_1)))$ 
11: end for
12: return  $\mathbf{X}_{\text{tr}} \in \mathbb{R}^{B \times N \times D}$ 

```

Together, the sequential use of ConvLSTM, GCN, and Transformer follows a coherent hierarchy of spatiotemporal feature extraction. ConvLSTM is placed first because it directly models short-term temporal dynamics while preserving the full spatial structure of the input. The GCN then refines these features by propagating information across local spatial neighborhoods, producing spatially consistent representations. Finally, the Transformer operates on these structured features to capture long-range spatial dependencies that cannot be modeled by convolution or graph aggregation alone. This progression from temporal encoding to local spatial integration to global attention ensures a logically grounded and efficient extraction of spatiotemporal patterns for high-resolution VI prediction.

3.5. Loss function

To train the model for pixel-wise image prediction, the Mean Squared Error (MSE) was used as the loss function. This choice is well suited for regression tasks involving continuous-valued outputs such as NDVI images, as it penalizes larger errors more heavily. Given the predicted image $\hat{Y} \in \mathbb{R}^{H \times W}$ and the ground truth image $Y \in \mathbb{R}^{H \times W}$, the MSE loss is computed over all pixels as:

$$\mathcal{L}_{\text{MSE}} = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W (\hat{Y}_{i,j} - Y_{i,j})^2 \quad (22)$$

This function ensures that the model is optimized to minimize the average squared error across all pixels in the output image. MSE is particularly effective in capturing overall structural accuracy and encouraging smoothness in spatially coherent predictions.

3.6. Evaluation metrics

To assess the performance of the model in pixel-wise NDVI prediction, multiple complementary evaluation metrics were employed, capturing both numerical accuracy and spatial fidelity. Specifically, Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), Nash-Sutcliffe Efficiency (NSE), Structural Similarity Index (SSIM), and Moran's I of residuals were used. These metrics are computed by comparing predicted and true NDVI values at each pixel location over the test set.

The Root Mean Squared Error (RMSE) measures the square root of the average squared differences between predicted and actual values,

giving more weight to larger errors:

$$\text{RMSE} = \sqrt{\frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W (\hat{Y}_{i,j} - Y_{i,j})^2} \quad (23)$$

The Mean Absolute Error (MAE) computes the average of the absolute differences, offering a robust and easily interpretable error estimate:

$$\text{MAE} = \frac{1}{H \times W} \sum_{i=1}^H \sum_{j=1}^W |\hat{Y}_{i,j} - Y_{i,j}| \quad (24)$$

The Nash–Sutcliffe Efficiency (NSE) evaluates predictive skill relative to the mean of the observations and is defined as:

$$\text{NSE} = 1 - \frac{\sum_{i,j} (Y_{i,j} - \hat{Y}_{i,j})^2}{\sum_{i,j} (Y_{i,j} - \bar{Y})^2} \quad (25)$$

where $\bar{Y}$ denotes the mean of the ground-truth NDVI values over all pixels. An NSE value close to 1 indicates high predictive skill, while values below 0 indicate performance worse than a mean-based predictor.

To explicitly evaluate spatial structural fidelity, the Structural Similarity Index (SSIM) was employed. SSIM measures similarity between predicted and reference NDVI maps by jointly considering luminance, contrast, and structural information:

$$\text{SSIM}(Y, \hat{Y}) = \frac{(2\mu_Y \mu_{\hat{Y}} + C_1)(2\sigma_{Y\hat{Y}} + C_2)}{(\mu_Y^2 + \mu_{\hat{Y}}^2 + C_1)(\sigma_Y^2 + \sigma_{\hat{Y}}^2 + C_2)} \quad (26)$$

where μ and σ denote mean and standard deviation, $\sigma_{Y\hat{Y}}$ is the covariance between predicted and observed NDVI, and C_1, C_2 are small constants for numerical stability. SSIM values range from 0 to 1, with values closer to 1 indicating higher structural similarity.

Finally, spatial autocorrelation of prediction errors was quantified using Moran's I computed on the residual field $R_{i,j} = \hat{Y}_{i,j} - Y_{i,j}$:

$$I = \frac{N}{W} \frac{\sum_p \sum_q w_{pq} (R_p - \bar{R})(R_q - \bar{R})}{\sum_p (R_p - \bar{R})^2} \quad (27)$$

where N is the number of pixels, w_{pq} denotes spatial adjacency weights, $W = \sum_{p,q} w_{pq}$, and $\bar{R}$ is the mean residual. Lower Moran's I values indicate reduced spatial clustering of errors, reflecting improved spatial consistency in predictions.

In addition, the computational complexity of each deep learning model was evaluated using the number of floating-point operations (FLOPs) and the total number of learnable parameters (Params). FLOPs provide a standard measure of computational cost by quantifying the number of floating-point operations required to process an input, with higher values indicating greater computational demand. The parameter count reflects the model's structural complexity and directly influences memory usage and storage requirements. Models with fewer parameters generally entail lower computational overhead and reduced memory consumption, thereby offering improved efficiency and scalability.

4. Datasets

This section describes the dataset used to train and evaluate the proposed architecture, comprising monthly single-channel images derived from Landsat satellite data, covering the Bizerte region in northern Tunisia from 1996 to 2025. The study area and data preparation processes are detailed below, highlighting the automated approach to data collection and preprocessing.

4.1. Study area

The study area, illustrated in Fig. 5, is located within Bizerte Governorate in northern Tunisia, geographically centered at latitude 37.25°N and longitude 9.80°E and bounded by the coordinates (9.70°E, 37.20°N, 9.90°E, 37.30°N). Bizerte Governorate covers a total surface area of

374,822 hectares and represents one of the most environmentally diverse regions in Tunisia, encompassing agricultural plains, forested mountainous areas, urban settlements, inland water bodies, and coastal zones connected to Lake Bizerte and the Mediterranean Sea.

According to official regional statistics (Data, 2025), agricultural land occupies 207,593 ha (55.4%) of the governorate, forests cover 99,135 ha (26.5%), rangelands account for 22,211 ha (5.9%), inland water bodies and lakes occupy 27,300 ha (7.3%), and non-cultivable and built-up areas represent 18,583 ha (5.0%). This heterogeneous land-cover composition results in pronounced spatial contrasts between adjacent parcels, particularly at the interfaces of croplands, forests, and urbanized zones. Agricultural areas exhibit strong seasonal NDVI variability driven by cropping cycles, while water bodies and urban surfaces introduce low-reflectance and mixed-pixel effects that complicate vegetation index reconstruction. The temporal coverage from 1996 to 2025, spanning Landsat 5, 7, and 8 missions, enables the analysis of both short-term phenological dynamics and long-term vegetation trends. This combination of spatial heterogeneity and long-term temporal depth provides a rigorous testbed for evaluating the proposed framework under realistic and complex environmental conditions.

4.2. Data collection and preprocessing

The dataset, detailed in Table 2, was generated through an automated pipeline implemented in Google Earth Engine (GEE), which processed Landsat 5 TM, Landsat 7 ETM+, and Landsat 8 OLI Surface Reflectance images spanning 1996–2025. These images, sourced from USGS Collection 2 Level-2, have a spatial resolution of 30 m and were cropped to a fixed spatial window of 372 × 743 pixels. The Collection 2 products provide standardized preprocessing including atmospheric correction (LEDAPS for Landsat 5–7, LaSRC for Landsat 8) and geometric correction (systematic and terrain-corrected L1TP), ensuring radiometric and spatial consistency across sensors and over time. Images were filtered using the Bizerte bounding box and a scene-level cloud cover threshold below 60%.

Cloud and shadow pixels were identified and masked using the QA_PIXEL quality band with the CFMask algorithm prior to NDVI computation. Landsat 7 ETM+ Scan Line Corrector-off (SLC-off) pixels were also flagged via the same quality band and treated as missing observations. A hybrid gap-filling strategy was uniformly applied to all missing data (clouds, shadows, SLC-off gaps): first, missing values were replaced with the maximum NDVI observed in the same calendar month across the time series to avoid underestimation of vegetation; second, a temporal smoothing step based on the inter-annual mean (year-1, year, year + 1) was applied to preserve phenological consistency; finally, persistent gaps were addressed using a 3 × 3 spatial mean filter, with a fallback to the global monthly mean where necessary. This approach ensured data continuity while minimizing interpolation artifacts.

The Normalized Difference Vegetation Index (NDVI) was computed as:

$$\text{NDVI} = \frac{\text{NIR} - \text{RED}}{\text{NIR} + \text{RED}} \quad (28)$$

using sensor-specific near-infrared (NIR) and red bands. To further harmonize cross-sensor differences, spectral band adjustments were applied to align the spectral response functions of Landsat 8 OLI with those of Landsat 5 TM and Landsat 7 ETM+. Residual sensor-induced noise was quantified over pseudo-invariant features (permanent bare soil and evergreen vegetation), revealing mean inter-sensor NDVI differences of 0.01–0.02 and standard deviations below 0.03, which are negligible relative to seasonal vegetation dynamics.

Although NDVI theoretically ranges from -1 to 1, values were linearly normalized to the [0,1] interval prior to model training. This normalization improves numerical stability during optimization, accelerates convergence, and aligns with the bounded output of the sigmoid activation

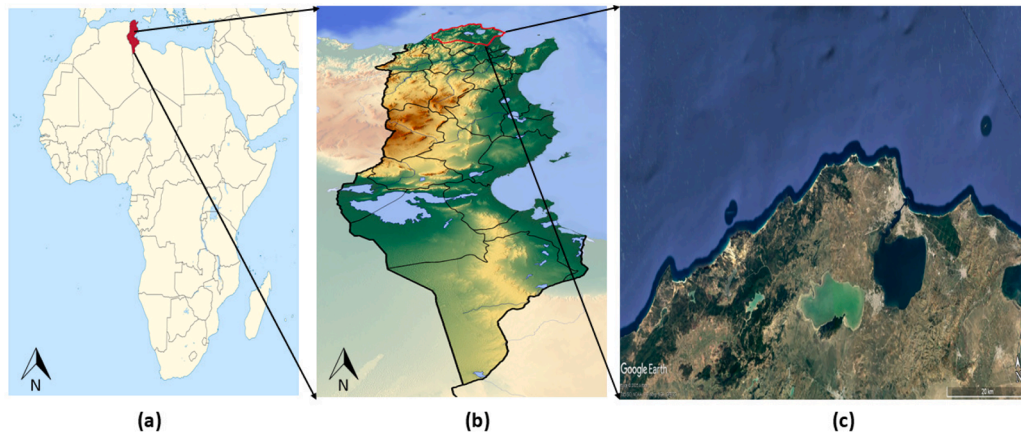


Fig. 5. Study area: (a) Geographic location of Tunisia in northern Africa, (b) Spatial extent of the study area (red boundary), and (c) Landsat scene of the study area.

Table 2
Input dataset characteristics.

Property	Description
Data Source	Landsat 5, 7, 8 Surface Reflectance (Google Earth Engine)
Temporal Coverage	1996–2024 (monthly)
Image Size	372 × 743 pixels
Resolution	30 m/pixel
Number of Images	348 (GeoTIFF)
Sequence Length	11-time steps
Data Type	Float32 (single-channel NDVI)
Memory Size before sequence creation	237 MB
Memory Size after sequence creation	3.49 GB
Preprocessing	Cloud/shadow masking, NDVI normalization missing values filling, median filtering, sequence creation

function used in the final layer. Crucially, negative NDVI values associated with non-vegetated surfaces are preserved through this invertible linear mapping rather than thresholding or exclusion, ensuring that full scene information is retained while maintaining compatibility with standard deep learning practices for remote sensing data (Ma et al., 2019; Yuan et al., 2020).

Images were arranged into sequences of 11 consecutive months to predict the NDVI of the following month, using an 80-20 train-test split. The 11-month input length was chosen based on the phenological characteristics of Mediterranean vegetation, which completes its annual growth cycle within this timeframe (Lobert et al., 2023; Zhang et al., 2003). This duration ensures observation of key phenological stages green-up, peak growth, senescence, and dormancy, thus providing the model with sufficient temporal context to learn seasonal vegetation dynamics while maintaining proximity to the prediction target. Shorter sequences risk omitting essential antecedent conditions from previous growing seasons, while longer sequences introduce diminishing informational returns due to strong annual periodicity in vegetation-climate relationships.

Table 3
Experimental hardware and software setup.

Locale Configuration		
Hardware	CPU	Intel i5-14400F
	GPU	GeForce RTX 5060
	RAM	32 GB
Software	System	Ubuntu 24.04
	Python	3.12.8
Environment	TensorFlow / Keras	2.17.0 / 3.4.1
	CUDA	12.5.0
	Cudnn	8.9.7

5. Experiments and results

5.1. Implementation details

The experimental implementation was conducted using a high-performance computing setup to ensure efficiency and stability throughout the training and testing processes. The hardware configuration included an Intel i5-14400F CPU, 32 GB of RAM, and an NVIDIA RTX 5060 GPU with 12 GB of VRAM, providing sufficient computational power for deep learning tasks. The experiments were carried out using TensorFlow/Keras as the primary framework, leveraging the CUDA environment to optimize GPU utilization (Table 3).

During training, we employed the Adam optimizer with an initial learning rate of 1×10^{-4} to balance convergence speed and stability, training the model for 100 epochs with a batch size of 2 to optimize memory usage and gradient updates. The process was guided by the MSE loss function and evaluated using RMSE, MAE, and NSE, to ensure a comprehensive performance assessment.

To enhance training stability and efficiency, we incorporated several callbacks:

- ModelCheckpoint saved the best model based on validation loss.
- ReduceLROnPlateau reduced the learning rate by a factor of 0.1 after 10 epochs without improvement (minimum: 1×10^{-7}).
- CSVLogger recorded detailed training metrics.
- TensorBoard provided real-time visualization.
- EarlyStopping halted training after 50 epochs without validation loss improvement to prevent overfitting and save computational resources.

These combined strategies contributed to a robust and efficient training process.

Table 4
Computational complexity and runtime comparison.

Model	FLOPs (G)	Params	Time /Epoch
ConvLSTM2D	38.5	261,409	8.35 min
ConvLSTM-3DCNN	44.3	223,297	18 min
ConvLSTM-GCN-Trans (Ours)	59.1	173,761	24.15 min

5.2. Computational complexity and runtime analysis

This subsection analyzes the computational efficiency of the proposed architecture in terms of model complexity, floating-point operations (FLOPs), and training runtime. The objective is to assess whether the integration of GCN and Transformer modules introduces prohibitive computational overhead, particularly given the batch size of 2 used throughout the experiments.

Table 4 compares representative models from the ablation study, including the baseline ConvLSTM2D, the ConvLSTM-3DCNN variant, and the proposed ConvLSTM-GCN-Transformer architecture. FLOPs are estimated for a single forward pass using the input configuration described in Table 3, while runtime corresponds to the average training time per epoch under identical training conditions.

As shown in Table 4, the proposed model achieves competitive computational efficiency, with FLOPs (59.1 G) and training time per epoch (24.15 min) remaining tractable despite its enhanced modeling capacity. Notably, the full model uses fewer parameters (173,761) than the ConvLSTM2D baseline, confirming that the performance gains are not achieved at the cost of excessive model size.

The moderate increase in runtime is primarily due to the sequential processing of ConvLSTM, GCN, and Transformer modules. Nevertheless, the design ensures that the computational overhead is contained: temporal dynamics are compactly encoded by ConvLSTM layers, local spatial dependencies are handled by sparse GCN operations, and the Transformer processes only low-dimensional feature embeddings (32 channels), thereby avoiding the quadratic cost typical of dense self-attention on raw pixels. This modular and efficient division of roles enables a favorable trade-off between expressiveness and efficiency, making the proposed architecture practical for operational Landsat NDVI forecasting even with constrained GPU memory.

5.3. Qualitative and quantitative analysis

5.3.1. Quantitative performance analysis

The quantitative performance of the proposed ConvLSTM-GCN-Transformer model was evaluated using multiple complementary metrics that assess numerical accuracy and spatial fidelity. As reported in Table 5, the proposed model achieves an RMSE of 0.034, an MAE of 0.017, and an NSE of 0.866, indicating high predictive accuracy and strong agreement with observed NDVI dynamics.

Beyond pointwise error metrics, the model attains a high Structural Similarity Index (SSIM = 0.863), confirming its ability to preserve spatial structures and texture patterns in predicted NDVI maps. In addition, the residual spatial autocorrelation, measured using Moran's I, is notably low (0.094), suggesting that remaining prediction errors exhibit minimal spatial organization and are largely free of systematic spatial artifacts.

These results demonstrate that the proposed architecture not only minimizes numerical errors but also produces spatially coherent and structurally consistent NDVI predictions.

5.3.2. Qualitative spatial analysis

Fig. 6 provides a qualitative comparison between observed and predicted NDVI maps, highlighting the spatial behavior of the proposed model across heterogeneous landscape types. The figure is organized into two rows, where the upper row shows the reference NDVI maps and the lower row shows the corresponding predictions. From left to

right, the columns display the full study area followed by zoomed-in regions representing forest landscapes, open agricultural fields, and urban vegetation zones.

The model demonstrates excellent performance in forested regions, where dense and continuous vegetation patterns are accurately preserved with minimal distortion. Spatial continuity and sharp boundaries are well maintained, indicating effective modeling of homogeneous high-NDVI areas. Similarly, predictions over open agricultural fields exhibit strong agreement with observations, capturing parcel-level variability and large-scale vegetation gradients, although minor smoothing effects are observed in areas with abrupt transitions.

In contrast, urban vegetation areas present greater predictive challenges due to their high spatial heterogeneity and mixed land-cover composition. In these regions, the predicted maps exhibit increased noise and localized discrepancies compared to the reference NDVI. This behavior reflects the intrinsic complexity of urban environments, where small vegetated patches are interwoven with impervious surfaces and exhibit rapid spatial variability.

Overall, these qualitative results confirm that the proposed ConvLSTM-GCN-Transformer architecture effectively captures both global vegetation patterns and local spatial structures. The strong visual consistency between predicted and observed NDVI maps, particularly in natural and agricultural landscapes, corroborates the quantitative findings and highlights the model's capacity to generalize across diverse land-cover types.

5.3.3. Uncertainty analysis

To enhance the credibility and interpretability of the proposed framework, an uncertainty analysis was conducted by examining the spatial and temporal distribution of prediction errors. Rather than introducing additional probabilistic assumptions, uncertainty is characterized through residual behavior across heterogeneous landscapes and seasonal dynamics, focusing on (i) spatial uncertainty and (ii) temporal uncertainty.

Spatial uncertainty is primarily observed in structurally complex regions, particularly urban and peri-urban vegetation zones. Quantitative analysis confirms this observation, as the residual spatial autocorrelation reaches its highest values in these regions, with Moran's I attaining approximately 0.18. As illustrated in Fig. 6, the model performs robustly in homogeneous landscapes such as forests and large agricultural parcels, where predicted NDVI maps exhibit strong spatial coherence and minimal noise. In contrast, urban vegetation areas display higher residual variability and localized artifacts. This behavior is attributed to sub-pixel land-cover heterogeneity, mixed spectral responses, and abrupt transitions between vegetated and built-up surfaces, which are difficult to represent accurately at Landsat resolution. Although residual spatial structure persists, the relatively low Moran's I values indicate that the remaining uncertainty reflects intrinsic scene complexity rather than systematic model bias or spatial over-smoothing.

Temporal uncertainty is assessed through the monthly evolution of prediction error over the test period, as shown in Fig. 7. The model achieves low and stable RMSE values during winter and early spring (December to April), corresponding to periods of relatively stable vegetation cover. A clear increase in uncertainty is observed during June and July, when RMSE reaches its maximum. This peak coincides with rapid vegetation dynamics in Mediterranean environments, particularly grain harvesting and post-harvest transitions, which induce abrupt NDVI changes over short time spans. These sharp temporal discontinuities challenge sequence-based prediction models, yet the uncertainty remains temporally localized and decreases again in late summer and autumn.

Overall, the uncertainty analysis demonstrates that elevated prediction errors are concentrated in specific spatial and temporal contexts characterized by high heterogeneity or abrupt dynamics. Importantly, uncertainty patterns are physically meaningful and consistent with known land-cover complexity and seasonal processes. This confirms that the proposed model delivers reliable predictions under most

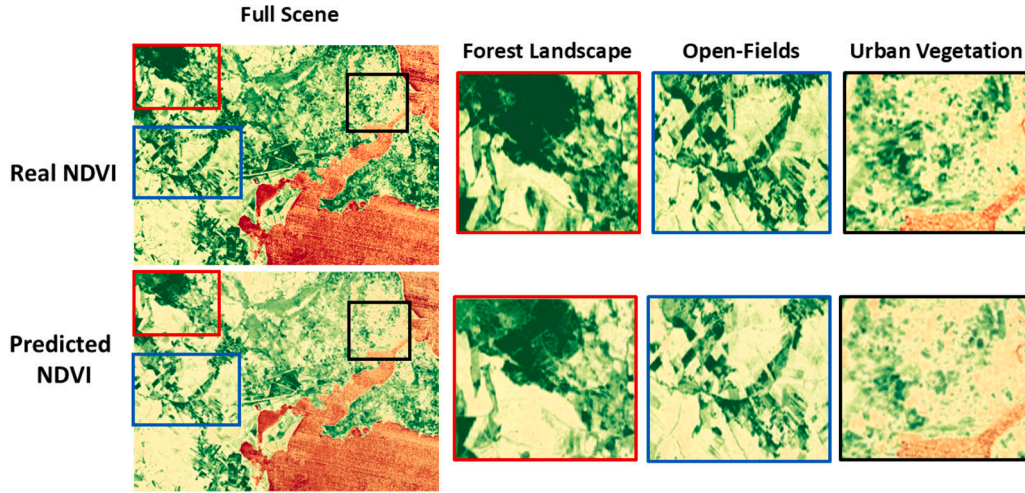


Fig. 6. Qualitative comparison between observed (top row) and predicted (bottom row) NDVI maps over the study area: Colored rectangles on the full-scene maps highlight the locations of the zoomed-in forest (red), open agricultural fields (blue), and urban vegetation (black) regions.

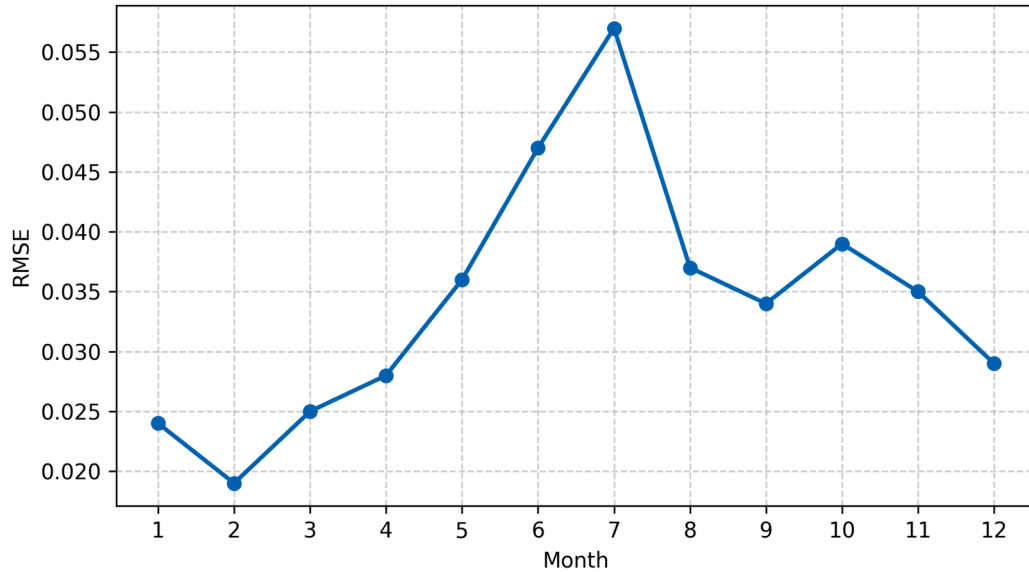


Fig. 7. Monthly RMSE of the proposed ConvLSTM-GCN-transformer model over the test set.

conditions while transparently exposing scenarios where intrinsic environmental variability limits predictive confidence.

5.4. Ablation study

5.4.1. Architectural ablation

A comprehensive ablation study was conducted to evaluate the individual and combined contributions of the ConvLSTM, GCN, and Transformer modules (Table 5). Five model variants were analyzed: (i) baseline ConvLSTM, (ii) ConvLSTM with 3DCNN, (iii) ConvLSTM with Transformer, (iv) ConvLSTM with GCN (4-neighbor), (v) ConvLSTM with GCN (8-neighbor), and (vi) full ConvLSTM-GCN-Transformer with 4-neighbor adjacency (Ours). Each ablation modifies only one component at a time while maintaining identical datasets, training configurations, hyperparameters, and random seeds to ensure a fair comparison. The Statistical significance of performance differences was verified using paired t-tests ($p < .05$).

The baseline ConvLSTM model serves as the reference, leveraging gated memory to capture short-term temporal dependencies and local convolutions for spatial processing. However, its performance remains

limited (RMSE = 0.161, NSE = 0.421), particularly in heterogeneous landscapes characterized by sharp land-cover transitions and complex textures. This limitation highlights the need for mechanisms capable of explicitly modeling structured spatial relationships.

Incorporating a 3DCNN block yields moderate improvements (RMSE = 0.131, NSE = 0.511). By treating the input as a spatiotemporal volume, the model improves spatial continuity and reduces speckle noise. Nevertheless, the lack of explicit spatial dependency modeling restricts its ability to preserve parcel-level structure in highly fragmented regions.

Integrating a Transformer module further enhances performance (RMSE = 0.105, NSE = 0.628) by enabling long-range contextual modeling through self-attention. This results in improved global consistency and smoother vegetation gradients. However, local spatial inconsistencies persist, as the Transformer alone does not enforce neighborhood-level spatial regularity.

The integration of a Graph Convolutional Network (GCN) introduces a substantial performance gain. Using a 4-neighbor adjacency graph (RMSE = 0.081, NSE = 0.691), the GCN explicitly enforces local spatial coherence, effectively suppressing isolated artifacts and improving

Table 5
Ablation study results.

Model	RMSE	NSE	MAE	SSIM	Moran's I	Params
ConvLSTM2D	0.161	0.421	0.206	0.691	0.257	261,409
ConvLSTM-3DCNN	0.131	0.511	0.162	0.714	0.244	223,297
ConvLSTM-Trans	0.105	0.628	0.113	0.750	0.190	184,321
ConvLSTM-GCN (4-neighbor)	0.081	0.691	0.093	0.811	0.126	152,641
ConvLSTM-GCN (8-neighbor)	0.078	0.717	0.086	0.823	0.112	152,641
ConvLSTM-GCN-Trans (4-neighbor) (Ours)	0.034	0.866	0.017	0.863	0.094	173,761

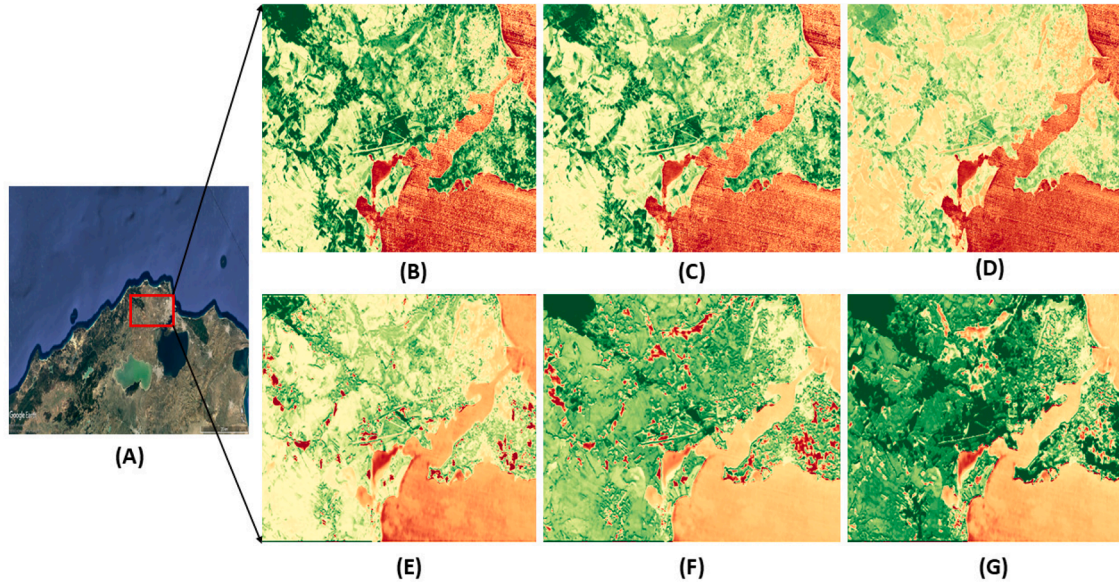


Fig. 8. Ablation study performed on a Landsat sub-scene: (A) Location of the selected sub-scene within the study area, shown with a red bounding box, (B) Ground-truth NDVI image, (C) ConvLSTM-GCN-Transformer prediction, (D) ConvLSTM-GCN , (E) ConvLSTM-Transformer, (F) ConvLSTM-3DCNN, and (G) ConvLSTM baseline.

homogeneity within agricultural parcels. This improvement is further confirmed by spatial fidelity metrics, with SSIM increasing to 0.811 and Moran's I of residuals decreasing to 0.126. This represents the most significant single-component improvement in spatial coherence metrics, demonstrating that explicit local spatial regularization through graph convolutions is crucial for capturing parcel-level structure.

To assess the sensitivity to graph connectivity, we evaluated an alternative 8-neighbor adjacency configuration under identical training conditions. While the denser graph yields modest improvements, it nearly doubles the number of edges (5.8M vs. 2.8M for 4-neighbor), increasing memory and computational costs. Importantly, the number of learnable parameters remains unchanged because the graph structure is fixed and does not contain trainable weights; only the GCN's weight matrix is learned. Since the denser connectivity does not yield proportional accuracy gains, we prioritize capturing long-range dependencies via the Transformer module, which addresses complementary spatial modeling needs more efficiently.

Therefore, for the full architecture, we retain the 4-neighbor configuration and integrate it with the Transformer. This hybrid approach achieves superior performance (RMSE = 0.034, SSIM = 0.863) by combining local spatial regularization (GCN) with global contextual modeling (Transformer), offering a more computationally efficient pathway to spatial coherence than graph densification alone. The visual results presented in Fig. 8 confirm that long-range modeling via attention provides greater marginal benefit than further refining local connectivity.

5.4.2. Sequence length ablation

The sensitivity of the model to temporal context length was evaluated through a sequence-length ablation study, testing input windows of 6, 9, 11, and 14 months under identical experimental conditions

Table 6
Sequence-length ablation study results.

Input Length (months)	RMSE	NSE	MAE	SSIM	Moran's I	GPU Use (GB)
6	0.096	0.679	0.091	0.782	0.149	4.2
9	0.071	0.783	0.062	0.817	0.116	6.9
11 (Selected)	0.034	0.866	0.017	0.863	0.094	9.5
14	0.031	0.873	0.021	0.874	0.085	11.8

(Table 6). Results indicate a clear trend: increasing temporal context improves predictive performance up to 11 months, beyond which returns diminish. The 6-month sequence yields the highest error (RMSE = 0.096), reflecting insufficient historical context to capture annual vegetation phenology. Performance improves at 9 months (RMSE = 0.071), but the 11-month configuration achieves the most favorable balance, with RMSE = 0.034, NSE = 0.866, and SSIM = 0.863. Although the 14-month sequence shows a slight RMSE reduction (0.031), it requires 24% more GPU memory (11.8 GB vs. 9.5 GB) while offering negligible gains in spatial coherence (Moran's I = 0.085). This suggests that 11 months effectively encapsulates a near-complete annual cycle, capturing essential seasonal vegetation dynamics while avoiding redundancy from older, less relevant observations, and maintaining computational tractability within the constraints of a 12 GB GPU with batch size 2.

5.4.3. Sensitivity analysis

We conducted a sensitivity analysis to evaluate the robustness of our model to different gap-filling strategies. Four methods were compared:

- Hybrid (Ours) – Our approach combines multiple steps: first, missing values due to clouds or artifacts are filled using the highest NDVI

Table 7
Sensitivity analysis of gap-filling methods on model performance.

Gap-Filling Method	RMSE	NSE	MAE	SSIM	Moran's I
Linear Interpolation	0.041	0.798	0.029	0.846	0.103
Savitzky-Golay	0.044	0.826	0.019	0.839	0.108
Spatio-Temporal Mean	0.052	0.709	0.041	0.825	0.120
Hybrid (Ours)	0.034	0.866	0.017	0.863	0.094

value observed per month, which provides a conservative estimate that avoids underestimation of vegetation activity. This is then refined through a temporal smoothing step based on the interannual mean (year-1, year, year + 1), which preserves phenological consistency across years. Finally, persistent gaps are addressed with a 3×3 spatial mean filter, and if necessary, a global monthly mean is used as a fallback. This hybrid method leverages both temporal persistence and spatial continuity to produce stable, gap-free NDVI sequences.

- (ii) Linear Interpolation – A simpler temporal-only approach where missing values are filled using linear interpolation between the nearest valid observations in the time series. This method assumes linear change between known points and does not incorporate spatial or multi-year information.
- (iii) Savitzky-Golay Smoothing – A temporal smoothing technique that fits a low-degree polynomial to successive windows of the time series (window size = 5, polynomial order = 2). This method effectively reduces high-frequency noise while preserving important phenological peaks and trends, but may smooth out abrupt changes.
- (iv) Spatio-Temporal Mean – A neighborhood-based approach where missing pixels are filled using the mean of valid pixels within a 3×3 spatial window across three consecutive time steps ($3 \times 3 \times 3$ cube). This method leverages both local spatial homogeneity and short-term temporal consistency.

As shown in Table 7, the proposed method achieved the best performance across all metrics, confirming the effectiveness of its multi-stage, spatially and temporally informed design. While Linear Interpolation and Savitzky-Golay produced competitive results, they exhibited slightly higher errors and lower spatial coherence. The Spatio-Temporal Mean method showed the weakest performance, likely due to over-smoothing of fine-scale vegetation patterns. Importantly, the relative performance ranking of the proposed architecture remained consistent across all gap-filling strategies, demonstrating that its advantages are not an artifact of preprocessing choices but rather stem from its architectural design.

In addition to our gap-filling sensitivity analysis, we conducted a comprehensive sensitivity study on NDVI normalization strategies. As shown in Table 8, we compared raw NDVI values in the $[-1, 1]$ range with tanh activation against our normalized $[0, 1]$ approach with sigmoid activation. The results demonstrate that our normalization strategy yields superior performance across all metrics: lower prediction errors, better spatial fidelity, and faster convergence (39 vs. 47 epochs). These findings confirm that normalization to $[0, 1]$ not only improves numerical stability but also enhances overall model accuracy and training efficiency.

5.5. Comparative study

To evaluate the performance of the proposed architecture, we conducted a comprehensive comparison with five representative state-of-the-art (SOTA) vegetation index forecasting methods. The baseline models include: the convolutional long short-term memory network (ConvLSTM) (Ahmad et al., 2023; Robin et al., 2022), which combines convolutional and recurrent layers for spatiotemporal modeling of NDVI sequences; Graph WaveNet (GWN) (Beyer et al., 2023), a graph-based architecture employing adaptive graph convolutions and gated temporal convolutions to capture spatial dependencies in non-Euclidean domains;

ConvLSTM-SAC-NL (Xu et al., 2024), an enhanced ConvLSTM variant that integrates spatial autocorrelation and non-local attention mechanisms; and the probabilistic crop growth model (PCGM) (Berger et al., 2019), a process-based approach that calibrates physiological parameters using historical growth data. These models were selected based on their methodological diversity, relevance to map-based NDVI forecasting, and the availability of published performance metrics under comparable experimental settings.

The quantitative comparison summarized in Table 9 shows that the proposed model achieves a substantially lower RMSE (0.034) than all considered baselines. This improvement reflects key architectural differences rather than solely increased model capacity. Compared to ConvLSTM-SAC-NL (Xu et al., 2024), which incorporates non-local attention without explicit graph-based spatial regularization, the proposed framework reduces RMSE by approximately 44%, indicating that combining local graph constraints with global attention more effectively preserves fine-scale spatial continuity. In comparison with Graph WaveNet (Beyer et al., 2023), which relies on adaptive graph learning, the fixed 4-neighbor GCN adopted here achieves comparable spatial fidelity while maintaining lower computational overhead, favoring scalability and stable performance for large-area vegetation monitoring. Although several reference studies report complementary indicators such as MAE, correlation coefficients, or NSE, RMSE remains the only metric consistently available across all baselines. To ensure methodological fairness and direct comparability, RMSE is therefore retained as the primary evaluation criterion.

Beyond aggregate error statistics, the proposed architecture demonstrates consistent behavior across heterogeneous landscape contexts within the study area. As illustrated in Fig. 6, the model preserves spatial structure and temporal coherence across contrasting land-cover types, including dense forests, open agricultural fields, and urban vegetation mosaics. These qualitative results complement the quantitative evaluation by confirming the model's ability to generalize across scenes characterized by distinct phenological patterns and spatial heterogeneity, while operating directly on full-resolution NDVI maps (743×372 pixels) without spatial downsampling.

5.6. Generalization study

Generalization is a cornerstone of model evaluation, ensuring that the proposed architecture can seamlessly adapt to new data sources and vegetation indices without retraining. This section presents two key experiments, sensor generalization and index generalization, that collectively demonstrate the architecture's robustness and flexibility in real-world applications.

5.6.1. Sensor generalization

To assess the robustness and transferability of our NDVI prediction model, we conducted a sensor generalization study by evaluating its performance on two independent vegetation index datasets: Sentinel-2 and MODIS (Table 10). These datasets differ significantly in spatial resolution, coverage, and sensor characteristics, providing a robust basis for evaluating cross-sensor generalization. Notably, the model was trained exclusively on Landsat NDVI maps at 30-m resolution, without any retraining or fine-tuning on Sentinel-2 or MODIS data. By applying the trained model directly to these new datasets, each resampled to the same spatial size (372×743 pixels), we aim to verify whether the spatiotemporal representations learned from Landsat imagery remain valid and effective across varying sensor characteristics, acquisition geometries, and spatial footprints.

The results, presented in Table 11, confirm the strong generalization capability of the proposed ConvLSTM-GCN-Transformer model across both Sentinel-2 and MODIS NDVI datasets. On Sentinel-2, the model achieves an RMSE of 0.067 and an NSE of 0.78, while on MODIS data, despite its coarser 250-m resolution, the model maintains a competitive RMSE of 0.09 and an impressive NSE of 0.905. Moreover, high SSIM

Table 8
NDVI normalization sensitivity analysis.

Strategy (activation)	RMSE	NSE	MAE	SSIM	Moran's I	Convergence
Raw NDVI [-1,1] (tanh)	0.039	0.824	0.036	0.833	0.102	47 epochs
Normalized NDVI [0,1] (sigmoid)	0.034	0.866	0.017	0.863	0.094	39 epochs

Table 9
Comparative evaluation against SOTA models.

Reference (Model)	RMSE
ConvLSTM (Robin et al., 2022)	0.188
ConvLSTM-SAC-NL (Xu et al., 2024)	0.061
ConvLSTM (Beyer et al., 2023)	0.068
Graph WaveNet (Beyer et al., 2023)	0.059
ConvLSTM (Ahmad et al., 2023)	0.078
PCGM (Berger et al., 2019)	0.098
Ours	0.034

Table 10
Specifications of the datasets used for cross-sensor generalization.

Sensor	Date Range	Spatial Coverage	Image Number(Shape)
Sentinel-2 (10-m)	2016–2024 (Monthly)	27.64 km ²	108(372 × 743)
MODIS (250-m)	2016–2024 (Monthly)	17,275 km ²	108(372 × 743)

Table 11
Performance metrics of the NDVI model on sentinel-2 and MODIS datasets.

Model	Sentinel-2 (NDVI)			MODIS 250-m (NDVI)		
	RMSE	NSE	SSIM	RMSE	NSE	SSIM
Ours	0.067	0.780	0.806	0.090	0.905	0.781

values of 0.806 for Sentinel-2 and 0.781 for MODIS indicate that the model preserves spatial structures and vegetation patterns consistently across sensors with markedly different resolutions. These results are particularly noteworthy given that the model was not retrained on these datasets, highlighting its ability to transfer learned representations to new spatial resolutions and sensor modalities, as shown in Figs. 9 and 10. This robust performance can be attributed to several architectural factors. The GCN block enables the model to learn spatial relationships independent of a fixed pixel grid, facilitating adaptation to different spatial granularities, and also the Transformer module, with its self-attention mechanism, captures scene-wide patterns regardless of local structure, making it inherently resilient to variations in resolution and image quality.

5.6.2. Index generalization

To evaluate the index-level generalization capability of the proposed architecture, we conducted a second experiment by training two separate models using the same methodological approach but with different vegetation indices: the Enhanced Vegetation Index (EVI) and the Soil-Adjusted Vegetation Index (SAVI), both derived from the Landsat time series. The equations for these indices are as follows:

$$EVI = 2.5 \times \frac{NIR - Red}{NIR + 6 \times Red - 7.5 \times Blue + 1} \quad (29)$$

$$SAVI = \frac{NIR - Red}{NIR + Red + L} \times (1 + L) \quad (30)$$

where L is a soil adjustment factor, typically set to 0.5.

Landsat was selected for its extensive temporal coverage from 1996 to 2025 and consistent 30-meter spatial resolution. Both models were trained and tested on the same study area (249 km²) with identical temporal and spatial splits. The Table 12 summarizes the dataset characteristics used in this experiment.

Table 12
Dataset characteristics for EVI and SAVI used in index generalization.

Index	Sensor	Spatial Coverage	Images Number (Shape)
EVI	Landsat (30-m)	249 km ²	348 (372 × 743)
SAVI	Landsat (30-m)	249 km ²	348 (372 × 743)

Table 13
Performance of the proposed architecture on EVI and SAVI prediction tasks.

Model	EVI			SAVI		
	RMSE	NSE	SSIM	RMSE	NSE	SSIM
ConvLSTM-GCN-Transformer	0.059	0.770	0.829	0.052	0.790	0.836

The numerical results (Table 13) highlight the robust generalization capability of the proposed architecture across different vegetation indices. Despite the shift from EVI to SAVI, the model maintains high predictive accuracy with minimal variation in error metrics. In addition to numerical accuracy, the model demonstrates high structural consistency, achieving an SSIM of 0.829 for EVI and 0.836 for SAVI, which reflects effective preservation of spatial patterns and vegetation structures in the predicted maps. This robustness stems from the architecture's ability to effectively model spatiotemporal dynamics and adapt to diverse data representations without requiring structural modifications. The training strategy, based on consistent preprocessing, data alignment, and temporal sampling, further supports this generalization. Specifically, the model achieved an RMSE of 0.059 and an NSE of 0.77 for EVI, and slightly improved to an RMSE of 0.052 and an NSE of 0.79 for SAVI. These results demonstrate the potential of the proposed method as a flexible and reliable solution for multiple vegetation indices, broadening its applicability across diverse remote sensing tasks.

6. Discussions

This study introduces a hybrid ConvLSTM-GCN-Transformer architecture for high-resolution, map-based vegetation index forecasting, designed to address complementary limitations of existing spatiotemporal models. The proposed framework integrates temporal memory (ConvLSTM), local spatial regularization (GCN), and global contextual modeling (Transformer), enabling coherent NDVI prediction across both space and time. On multi-decadal Landsat NDVI data, the model achieves an RMSE of 0.034 and an NSE of 0.866, outperforming convolutional, graph-based, and attention-based baselines. Beyond numerical accuracy, the model exhibits high spatial fidelity (SSIM = 0.863) and low residual spatial autocorrelation (Moran's I = 0.094), indicating that improvements extend beyond point-wise error reduction to the preservation of spatial structure. These results emphasize the importance of jointly modeling temporal dynamics, local spatial continuity, and long-range dependencies for structurally consistent vegetation index forecasting in heterogeneous landscapes.

The ablation study provides insight into the performance advantages of the proposed architecture relative to individual baselines. The ConvLSTM2D baseline effectively captures short-term temporal dynamics but remains constrained by its local convolutional receptive field, resulting in spatially noisy predictions (SSIM = 0.691, RMSE = 0.161). Incorporating a GCN directly addresses this limitation by explicitly enforcing spatial consistency through graph-based propagation, leading to the

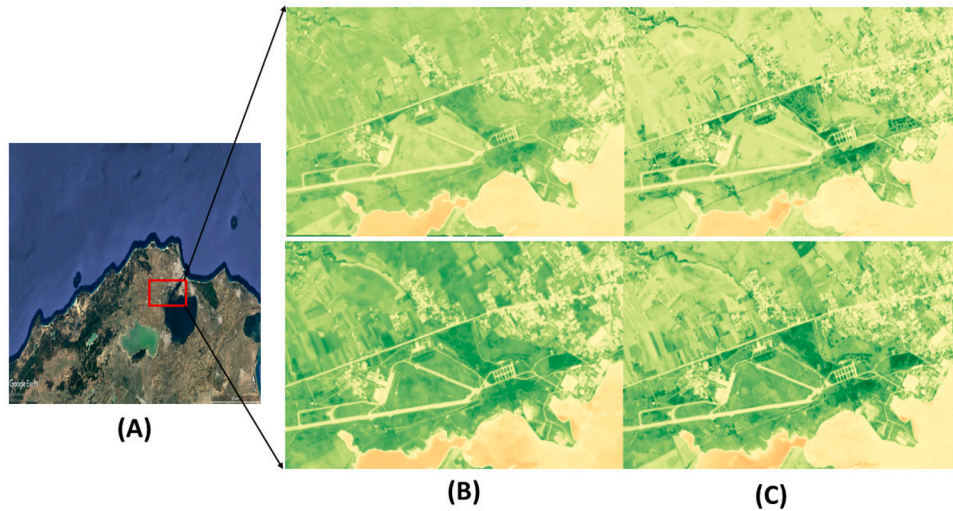


Fig. 9. NDVI prediction results on Sentinel-2 data: (A) Location of the selected sub-scene within the study area, shown with a red bounding box, (B) Ground-truth NDVI image, and (C) Predicted NDVI image generated by the proposed ConvLSTM-GCN-Transformer model.

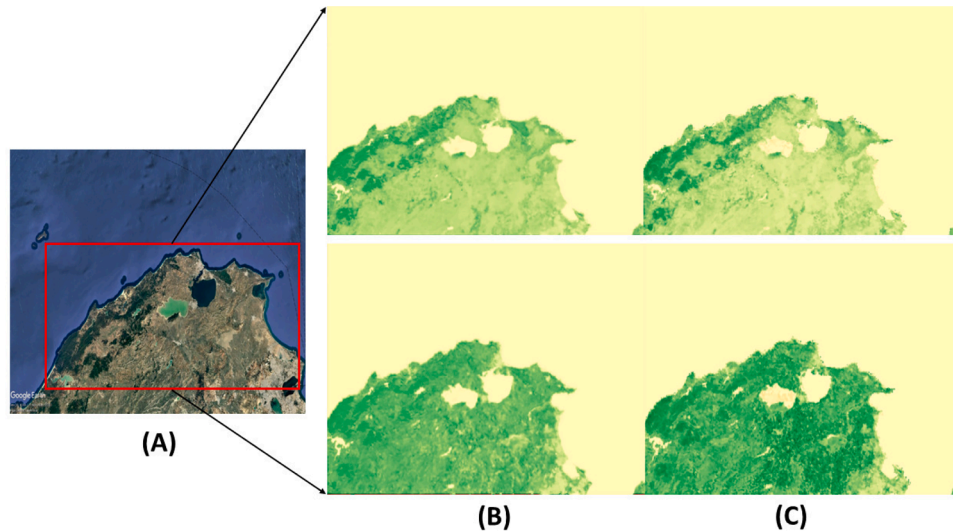


Fig. 10. NDVI prediction results on MODIS (250-m) data: (A) Location of the selected sub-scene within the study area, shown with a red bounding box, (B) Ground-truth NDVI image, and (C) Predicted NDVI image generated by the proposed model.

most pronounced single-component improvement. The 4-neighbor GCN configuration increases SSIM by 0.06 compared to the Transformer-only variant, demonstrating that modeling local adjacency is more effective than increasing temporal depth or contextual complexity alone. Extending adjacency to 8-neighbors yields only marginal computational gains despite a substantial increase in graph density, highlighting a clear trade-off between spatial refinement and cost at 30-m resolution. Compared to 3D CNN approaches, the GCN offers greater flexibility by decoupling spatial relationships from fixed kernel structures, making it better suited to irregular parcel geometries and heterogeneous land-cover patterns.

The Transformer component complements the GCN by addressing a different limitation inherent to convolutional and graph-based operators, namely their restricted spatial reach. While the GCN enhances local smoothness, it cannot align phenological patterns across distant regions. Self-attention enables the model to capture long-range dependencies and synchronize vegetation dynamics over spatially separated but ecologically similar areas. This capability explains the additional reduction in residual spatial autocorrelation and the improvement in SSIM beyond what is achievable through local regularization alone. Im-

portantly, the Transformer operates on features already structured by the GCN, ensuring that global context is incorporated without degrading local spatial coherence.

In addition to the ablation analysis, comparison with external state-of-the-art models further contextualizes the observed performance gains. Compared to ConvLSTM-SAC-NL, which integrates non-local attention but lacks explicit graph-based spatial regularization, the proposed framework achieves a 44% lower RMSE (0.034 vs. 0.061). This indicates that combining local graph constraints with global attention is more effective than attention alone for preserving parcel boundaries in agricultural landscapes. In comparison with Graph WaveNet, which relies on adaptive graph learning, the fixed 4-neighbor GCN adopted here achieves comparable spatial fidelity with lower computational overhead, favoring scalability and stable performance for large-area vegetation monitoring.

Sensitivity analyses further clarify the robustness of the proposed framework under varying temporal and preprocessing conditions. An input length of 11 months yields optimal performance, as it captures a complete Mediterranean phenological cycle and provides sufficient temporal context for learning seasonal dynamics without introducing

redundant information. Shorter sequences fail to represent full growth-senescence transitions, while longer sequences increase memory and computational cost without measurable accuracy gains. The model also shows stable relative performance across different gap-filling strategies, indicating limited sensitivity to interpolation artifacts. This robustness arises from the complementary design of the architecture: temporal continuity enforced by the ConvLSTM mitigates noise from missing observations, the GCN regularizes local spatial inconsistencies, and the Transformer integrates global context to compensate for residual distortions. Normalizing NDVI values to the $[0, 1]$ range improves numerical stability by ensuring scale consistency between input features and the sigmoid-constrained output, reducing gradient saturation and preventing error amplification during backpropagation, while preserving the relative amplitude of vegetation dynamics.

From a computational perspective, the proposed architecture balances expressiveness and efficiency despite incorporating pixel-level self-attention. Although attention mechanisms exhibit quadratic complexity with respect to token count, this cost is mitigated by applying self-attention only after temporal and spatial dimensionality reduction. The ConvLSTM and GCN compress representations into compact 32-channel embeddings, significantly reducing per-token computation. Consequently, the model enables global attention over approximately 276k spatial tokens without resorting to patching or sparse approximations, while maintaining practical efficiency. The reported FLOPs (59.1G) and per-epoch training time (24.15 minutes) remain comparable to convolutional baselines, justifying the added complexity through measurable gains in spatial coherence and predictive reliability.

Analysis of prediction errors shows that residual inaccuracies are primarily concentrated in spatially fragmented environments, such as urban-vegetation interfaces and mixed land-cover mosaics. In these areas, abrupt transitions and fine-scale heterogeneity challenge both local graph smoothness assumptions and global contextual alignment. Nevertheless, these residual errors affect a limited portion of the spatial domain and have a reduced impact on large-scale vegetation monitoring applications, which typically rely on spatially aggregated indicators rather than individual pixels. This is reflected in the overall low Moran's I value (0.094), although moderate values (≈ 0.18) are observed in highly heterogeneous urban vegetation zones.

The framework also demonstrates strong cross-sensor and cross-index generalization. Although trained exclusively on 30m Landsat data, the model transfers effectively to Sentinel-2 (10-m) and MODIS (250-m) without retraining. This suggests that the spatial relationships learned by the 4-neighbor GCN remain meaningful across moderate-to-coarse spatial resolutions. Similarly, the successful prediction of EVI and SAVI using the same architecture indicates that the model captures underlying vegetation dynamics rather than index-specific spectral properties. Such adaptability is important in operational contexts where sensor availability and vegetation indices may vary over time.

Finally, the proposed framework is well aligned with real-world operational applications, such as drought early-warning systems. In these scenarios, fixed NDVI thresholds (e.g., $\text{NDVI} < 0.35$) directly trigger management actions with significant economic and environmental implications. False positives can lead to unnecessary resource allocation, while false negatives may result in delayed responses to emerging drought conditions. The improved spatial coherence and reduced residual autocorrelation achieved by the model support more reliable threshold-based decision-making. At the same time, spatial variability in prediction reliability, particularly in fragmented urban vegetation, highlights the importance of accounting for uncertainty when deploying such systems at scale.

However, the current framework presents certain limitations related to complex land-cover fragmentation and uncertainty characterization. These limitations are discussed in detail in the following section.

7. Limitations and future works

Despite the strong predictive performance and cross-sensor generalization demonstrated by the proposed framework, several limitations remain that suggest directions for future research. The current spatial graph construction relies on a fixed local adjacency scheme, which is well suited to medium and coarse spatial resolutions but may become overly simplistic when applied to very high-resolution imagery. In scenarios involving airborne data or high-resolution satellite images, fine-scale spatial structures, sharp object boundaries, and heterogeneous land-cover patterns require richer spatial representations. Future work should therefore investigate graph densification strategies, such as expanded neighborhood connectivity or adaptive graph learning mechanisms, which can dynamically model spatial relationships and are expected to better capture detailed spatial interactions at finer resolutions.

Another limitation concerns the absence of explicit positional encoding within the Transformer component. Although spatial dependencies are implicitly modeled through convolutional operations and graph connectivity, explicit absolute or relative positional embeddings could further enhance spatial awareness, particularly in irregular graph structures or multi-resolution configurations. Incorporating positional encoding represents a natural extension of the current architecture and may improve the modeling of long-range spatial dependencies and global contextual relationships.

The proposed framework uses a fixed-length temporal input sequence, which effectively captures vegetation dynamics while minimizing computational overhead. This allows the model to represent key phenological patterns and seasonal transitions without excessive memory or runtime demands, making it scalable for large-scale remote sensing. While suitable for regional and decadal monitoring, applications like long-term ecosystem change, climate-driven trends, or extreme-event analysis may benefit from longer temporal contexts, as noted in recent long-horizon vegetation studies. Extending to longer sequences is thus a natural future direction, particularly where high-performance computing is available.

Finally, while an initial uncertainty analysis is introduced through the examination of spatial residual structure and temporal error variability, comprehensive uncertainty quantification remains an open research direction. More advanced methodologies, including ensemble-based approaches, probabilistic modeling, or Bayesian deep learning frameworks, could provide pixel-wise confidence estimates and improve the interpretability and reliability of predictions. Such extensions would further strengthen the applicability of the proposed framework for operational remote sensing, environmental monitoring, and decision-support systems.

8. Conclusion

This study proposes a novel hybrid spatiotemporal framework for long-term NDVI prediction that integrates ConvLSTM, Graph Convolutional Networks (GCN), and Transformer architectures. The proposed model is designed to jointly capture temporal dynamics, local spatial coherence, and long-range contextual dependencies in multi-decadal Landsat time-series data, while maintaining computational feasibility for large-scale applications.

Comprehensive experiments, including ablation analysis, cross-sensor evaluation, and multi-index prediction tasks, were conducted to assess the effectiveness of each architectural component. The results demonstrate that the proposed ConvLSTM-GCN-Transformer architecture consistently outperforms baseline models in terms of numerical accuracy (RMSE, MAE, NSE) and spatial fidelity (SSIM and residual Moran's I). In particular, the integration of sparse graph-based spatial modeling significantly improves local spatial coherence, while the Transformer module enhances the representation of scene-level contextual patterns without incurring excessive computational cost. The proposed framework exhibits strong robustness across heterogeneous

landscapes, temporal regimes, and sensor modalities. High performance is maintained when transferring learned representations to Sentinel-2 and MODIS datasets, as well as when predicting alternative vegetation indices such as EVI and SAVI, without architectural modification or re-training. Furthermore, the uncertainty analysis confirms that elevated prediction errors are confined to physically meaningful scenarios, such as urban vegetation heterogeneity and periods of abrupt seasonal transitions, reinforcing the credibility of the model outputs.

Overall, this work demonstrates that combining structured spatial priors with temporal modeling and global attention provides an effective and scalable solution for regional-scale vegetation monitoring. The proposed architecture offers a reliable foundation for long-term environmental analysis and supports future applications in ecosystem monitoring, agricultural assessment, and climate-driven vegetation dynamics.

CRediT authorship contribution statement

Mohamed Chahine Bouaziz: Conceptualization, Methodology, Software, Writing – original draft; **Ali Ben Abbes:** Supervision, Validation, Formal analysis; **Mourad El Koundi:** Supervision, Validation, Data curation; **Imed Riadh Farah:** Supervision, Project administration, Funding acquisition.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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